

Olympiad Test Paper — Mathematics (Class 7) — Paper 4

Chapter 10: Operations with Integers

Paper 4 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	10 — Operations with Integers
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. National-tier signed-integer reasoning: long add/subtract chains, deeply nested brackets under BODMAS, parity-of-negative-factors arguments, distributive manipulations, and word problems where the classic slips (subtracting a negative ADDS, two negatives MULTIPLY positive, $(-2)^3 \neq -2^3$, a hidden 0 collapsing a product) are baked into the distractors. Do not write on this paper — mark answers on your answer sheet.

1. A counter at 0 on a number line is moved +9, then -14, then -7, then +5, then -1. Its final position is:

- (A) -8 (B) +8
(C) -12 (D) +36

2. Compute $(-17) + (+23) + (-31) + (+12) + (-6)$.

- (A) -45 (B) -19
(C) -7 (D) +89

3. Which single statement about integer operations is FALSE?

- (A) The integers are closed under subtraction.
(B) Subtraction is not commutative in general: $a - b$ need not equal $b - a$.
(C) Subtraction of integers is associative: $(a - b) - c$ always equals $a - (b - c)$.
(D) Adding the opposite of b to a gives the same result as $a - b$.

4. Evaluate $(-120) \div (-5) \div (-6) \div (+2)$.

- (A) +2 (B) -2
(C) -4 (D) +4

5. Evaluate $(-6) + (-4) \times (-5) - (-3) \times (+2)$.

(A) -12

(B) 20

(C) 26

(D) 8

6. Which expression equals 0 purely by the additive-inverse property, with no magnitude arithmetic?

(A) $(-47) \times 0$

(B) $47 - (-47)$

(C) $(-47) + 47$

(D) $(-47) + (-47)$

7. For any integer a , the expression $a \times (-1) \times (-1) \times (-1)$ equals:

(A) a itself

(B) always negative

(C) the opposite of a

(D) always 0

8. At 5 a.m. the temperature was -11°C . It then fell 3°C , rose 8°C , and fell 6°C . The temperature now is:

(A) -10°C

(B) -28°C

(C) $+4^{\circ}\text{C}$

(D) -12°C

9. Compute the product $(-2) \times (-3) \times (-1) \times (-4) \times (-5)$.

(A) -120

(B) +120

(C) -15

(D) +15

10. A drone starts at altitude 0 m, climbs 15 m, descends 23 m, descends 8 m, then climbs 11 m. Its final altitude is:

(A) -5 m

(B) +5 m

(C) -45 m

(D) +57 m

11. Which sign-rule conclusion is INCORRECT?

(A) A negative integer times a positive integer gives a negative product.

(B) A negative integer minus a larger-magnitude negative integer can give a positive result.

(C) The product of three negative integers is negative.

(D) A negative integer divided by a negative integer gives a negative quotient.

12. On a number line you start at +4, move 6 left, then 9 left, then 7 right, then 8 left, then 3 right. Where do you land?

(A) -9

(B) -15

(C) +9

(D) -5

13. Evaluate $(-3) \times [(-5) + (-2) - (-4)] + (-6)$.

(A) +3

(B) -15

(C) +21

(D) -9

14. The product of seven non-zero integers is negative. The number of negative factors among them must be:

- (A) even (B) exactly seven
(C) exactly one (D) odd

15. Evaluate $(-84) \div (-7) + (-5) \times (-3) - (-2) \times (+4)$.

- (A) 19 (B) 35
(C) -11 (D) 27

16. By how much is (-13) greater than (-41) ?

- (A) -28 (B) 54
(C) 28 (D) -54

17. Evaluate $(-2)^4 + (-2)^3 - (-2)^2$.

- (A) -4 (B) 28
(C) 4 (D) -12

18. Evaluate $30 - 48 \div (-6) \times 2 + (-5)$.

- (A) 41 (B) 9
(C) 25 (D) -9

19. Which statement is TRUE about the role of 1 and 0 for integers?

- (A) 0 is the additive identity and 1 is the multiplicative identity for every integer.
(B) 1 is the additive identity and 0 is the multiplicative identity.
(C) Multiplying any integer by 0 gives that integer back.
(D) 0 is its own opposite, but it has no role as an identity.

20. Compute $(-1) \times (-2) \times (-3) \times (-4) \times (-5) \times (-6)$.

- (A) +720 (B) -720
(C) +21 (D) -21

21. A submarine at -540 m rises in equal steps and after 6 steps reaches -60 m. Each step was:

- (A) -80 m (B) $+80$ m
(C) $+90$ m (D) $+100$ m

22. For any integer n , $n \times (-1) \times n \times (-1)$ equals:

- (A) $-n^2$ (B) n^2
(C) $2n$ (D) always negative

23. Compute $(-12) + (-12) + (-12) + (-12) + (-12) + (-12) + (-12)$.

- (A) -72 (B) -96
(C) +84 (D) -84

24. Evaluate $(-90) \div [(-4) + (-1)] - (-6)$.

- (A) +12 (B) +24
(C) -24 (D) +18

25. Which property justifies rewriting $(-9) \times (15 + (-6))$ as $(-9) \times 15 + (-9) \times (-6)$?

- (A) associative property of multiplication
- (B) commutative property of addition
- (C) additive inverse property
- (D) distributive property of multiplication over addition

26. Evaluate $(-144) \div 6 \times (-1) \div (-3) + (-4)$.

- (A) -12
- (B) -4
- (C) 12
- (D) -20

27. In a contest, +5 marks are given for each correct answer, -3 for each wrong one, and -1 for each unattempted. Of 20 questions a student gets 12 right, 5 wrong, and leaves 3 unattempted. The total score is:

- (A) 45
- (B) 60
- (C) 42
- (D) 48

28. Which value makes $(-7) \times \underline{\hspace{1cm}} + 11 = -31$ true?

- (A) -6
- (B) -3
- (C) 3
- (D) 6

29. Evaluate $(-5) - [(-8) - (-3) + (-2)]$.

- (A) -12
- (B) +12
- (C) +2
- (D) -2

30. Evaluate $[(-6) - (-10)] \times [(-3) - (+5)]$.

- (A) +32
- (B) -16
- (C) +16
- (D) -32

31. The sum of an integer and its opposite is then subtracted from (-29) . The result is:

- (A) 0
- (B) +29
- (C) -29
- (D) -58

32. On a number line, order these from least to greatest: $(-3) \times (-2)$, $(-15) \div 3$, $(-4) + (-4)$, $6 - 9$.

- (A) $(-15) \div 3$, $(-4) + (-4)$, $6 - 9$, $(-3) \times (-2)$
- (B) $(-3) \times (-2)$, $6 - 9$, $(-15) \div 3$, $(-4) + (-4)$
- (C) $(-4) + (-4)$, $6 - 9$, $(-15) \div 3$, $(-3) \times (-2)$
- (D) $(-4) + (-4)$, $(-15) \div 3$, $6 - 9$, $(-3) \times (-2)$

33. Evaluate $(-4) \times (-4) - (-4) - 4 \times (-4)$.

- (A) +4
- (B) +36
- (C) +28
- (D) -4

34. If p, q, r are all negative integers, then $p \times q \times r \times (p + q + r)$ is:

- (A) positive
- (B) negative
- (C) zero
- (D) cannot be determined

- 35.** Evaluate $(14 - 30) \times (-3) + (-56) \div (-7)$.
(A) 40 (B) -40
(C) 56 (D) -56
- 36.** The product $(-2) \times (-4) \times (-6) \times (-8) \times (-10)$ (the first five even negatives) is:
(A) positive (B) zero
(C) -30 (D) negative
- 37.** Which value makes $(-96) \div \underline{\hspace{1cm}} = +8$ true?
(A) +12 (B) -88
(C) +88 (D) -12
- 38.** Using a distributive step, $(-6) \times 98$ is best written as:
(A) $(-6) \times 100 + (-6) \times 2 = -600 - 12 = -612$
(B) $(-6) \times 100 + (-6) \times (-2) = -600 + 12 = -588$
(C) $(-6) \times 100 - (-6) \times 2 = -600 + 12 = -588$ but equals -612
(D) $(-6) \times 100 + 6 \times (-2) = -612$
- 39.** The product of two integers is +56 and one of them is -7. The other integer is:
(A) -8 (B) +8
(C) -392 (D) +392
- 40.** A trekker's elevation changes by -250 m, +180 m, -90 m, +60 m during a descent-heavy hike. Three times this net change equals:
(A) -100 m (B) +300 m
(C) -300 m (D) -580 m
- 41.** Which list of integers is in DECREASING (descending) order?
(A) -15, -8, -3, 0, 2, 7 (B) 7, 2, -3, 0, -8, -15
(C) 7, 2, 0, -3, -8, -15 (D) 7, 0, 2, -3, -15, -8
- 42.** Evaluate $(-7) \times [(-3) - (-8)] - (-7) \times (-4)$.
(A) +7 (B) -7
(C) +63 (D) -63
- 43.** The integers are closed under which collection of operations, and not under the one omitted?
(A) closed under addition, subtraction, and multiplication; not under division
(B) closed under all four of addition, subtraction, multiplication, and division
(C) closed under addition and multiplication only; not under subtraction
(D) closed under division and multiplication only
- 44.** Evaluate $(-2)^3 \times (-3)^2 - (-4)^2 \div (-2)$.
(A) -80 (B) -64
(C) -72 (D) +64

- 45.** Two integers have a sum of -7 and a product of $+12$. The integers are:
 (A) 3 and 4 (B) -6 and -1
 (C) -12 and -1 (D) -3 and -4
- 46.** Evaluate $(-13) - (-21) - (+8) - (-6) - (+4)$.
 (A) -2 (B) -52
 (C) $+10$ (D) $+2$
- 47.** Which expression equals the SMALLEST (most negative) integer?
 (A) $(-3) \times (-3) \times (-3)$ (B) $(-2) \times (-2) \times (-2) \times (-2) \times (-2)$
 (C) $(-8) - (-2)$ (D) $(-10) + (-4)$
- 48.** Evaluate $(-128) \div (-8) \div (-2) - (-3) \times (-2)$.
 (A) -2 (B) -14
 (C) $+14$ (D) $+22$
- 49.** A bank account at ₹0 records a debit of ₹320, a credit of ₹185, a debit of ₹95, and a credit of ₹430. The closing balance is:
 (A) $-\text{₹}200$ (B) $+\text{₹}200$
 (C) $+\text{₹}1030$ (D) $-\text{₹}230$
- 50.** Evaluate $(-9) - 6 \times (-4) + (-48) \div (-8) - 5$.
 (A) -16 (B) 10
 (C) 16 (D) 22
- 51.** A lift starts at floor $+5$ (basements are negative). It goes down 9, up 4, down 7, up 2, then down 3 floors. It ends at floor:
 (A) -18 (B) $+8$
 (C) -8 (D) -2
- 52.** Evaluate $(-15) \times (-4) - (-15) \times (+2) + (-15) \times (-1)$.
 (A) 105 (B) 75
 (C) 45 (D) -105
- 53.** Evaluate $-7^2 + (-7)^2 + (-1)^3$.
 (A) $+97$ (B) -97
 (C) -1 (D) $+1$
- 54.** Evaluate $[(-4) \times (-3) - 2] \times [(-2) + (-1) - (-4)]$.
 (A) $+10$ (B) -10
 (C) $+30$ (D) -30
- 55.** Evaluate $(-6) \times (-5) + (-6) \times (-3) - (-6) \times (+2)$.
 (A) -60 (B) 12
 (C) 60 (D) 48

56. The product of an odd number of negative integers and one factor of 0 (all other factors non-zero) is:

- (A) negative
- (B) zero
- (C) positive
- (D) depends on the count of negatives

57. A weather station logs hourly temperature changes of -2 , -2 , $+5$, -3 , -2 , $+1$ °C starting from 4 °C. The final temperature is:

- (A) -3 °C
- (B) $+7$ °C
- (C) -1 °C
- (D) $+1$ °C

58. Evaluate $(-10) - \{(-4) - [(-6) + (-2)]\}$.

- (A) -2
- (B) $+14$
- (C) -6
- (D) -14

59. The product of an EVEN number of negative integers together with several positive integers (none zero) is always:

- (A) negative
- (B) positive
- (C) zero
- (D) even

60. Evaluate $[(-7) \times (-2) - 8] \div [(-1) + (-2)] \times (-3)$.

- (A) $+6$
- (B) -6
- (C) $+18$
- (D) -2

Solutions — Mathematics (Class 7), Chapter 10: Operations with Integers — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	D	21	B	31	C	41	C	51	C
2	B	12	A	22	B	32	D	42	D	52	A
3	C	13	A	23	D	33	B	43	A	53	C
4	B	14	D	24	B	34	A	44	B	54	A
5	B	15	B	25	D	35	C	45	D	55	C
6	C	16	C	26	A	36	D	46	D	56	B
7	C	17	C	27	C	37	D	47	B	57	D
8	D	18	A	28	D	38	B	48	B	58	D
9	A	19	A	29	C	39	A	49	B	59	B
10	A	20	A	30	D	40	C	50	C	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) Add the signed steps: $9 + (-14) = -5$; $-5 + (-7) = -12$; $-12 + 5 = -7$; $-7 + (-1) = -8$. Treating every step as a magnitude to add ($9+14+7+5+1$) gives 36; dropping a sign on the -1 leaves $-7-1$ mishandled as -12; flipping the final result's sign gives +8.

2. (B) Positives: $23 + 12 = 35$. Negatives: $17 + 31 + 6 = 54$. $35 - 54 = -19$. Summing all magnitudes gives 89; adding only the three negatives gives -54 against +35 mis-paired as -45; mis-grouping +12 as a negative yields -7.

3. (C) Associativity fails for subtraction: take $a=10$, $b=4$, $c=3 \rightarrow (10-4)-3 = 3$ but $10-(4-3) = 9$, so the claim it 'always' holds is false. Closure holds (any integer minus an integer is an integer), non-commutativity is true, and $a - b = a + (-b)$ is the defining identity.

4. (B) Left to right: $(-120) \div (-5) = +24$; $+24 \div (-6) = -4$; $-4 \div 2 = -2$. There are three negative signs among the divisions' inputs (the -5 and -6 give one positive, then dividing

into -6 once more is already counted) — concretely the running quotient is $+24, -4, -2$; an even count of negatives would force $+2$.

5. (B) Multiply first: $(-4) \times (-5) = +20$ and $(-3) \times (+2) = -6$. Expression becomes $(-6) + 20 - (-6) = -6 + 20 + 6 = 20$. Adding left to right before multiplying gives wrong magnitudes; reading $-(-6)$ as -6 gives 8; sign-slipping the $+20$ to -12 region comes from $(-4) \times (-5)$ taken as -20 .

6. (C) An integer plus its opposite is 0 by the additive-inverse property: $(-47) + 47 = 0$. $(-47) \times 0 = 0$ but uses the zero-product (multiplicative) property, not additive inverse; $47 - (-47) = 94$; $(-47) + (-47) = -94$.

7. (C) Three factors of -1 multiply to $(-1)^3 = -1$, so $a \times (-1) = -a$, the opposite of a . An even count of -1 factors would return a itself; the sign of $-a$ depends on a (e.g. $a = -5$ gives $+5$), so it is not 'always negative'.

8. (D) Track signed changes from -11 : $-11 - 3 = -14$; $-14 + 8 = -6$; $-6 - 6 = -12$. Reading the first 'fell 3' as a rise gives -10 ; adding every magnitude to -11 gives -28 ; ignoring the start and netting the changes $(-3+8-6=-1)$ misplaces it.

9. (A) Magnitude: $2 \times 3 \times 1 \times 4 \times 5 = 120$. There are five negative factors; an odd count makes the product negative, so -120 . An even count would give $+120$; adding the magnitudes gives 15.

10. (A) Signed sum: $0 + 15 - 23 - 8 + 11 = -5$. Adding all four magnitudes gives 57; taking only the descents $(-23-8)$ against $+15$ mis-pairs to $-16/-45$; flipping the result sign gives $+5$.

11. (D) Negative $\div$ negative is positive (e.g. $(-12) \div (-4) = +3$), so calling it negative is wrong. Negative $\times$ positive is negative; $(-4) - (-9) = +5$ is positive; an odd count of three negatives gives a negative product.

12. (A) Each 'left' subtracts, each 'right' adds: $4 - 6 - 9 + 7 - 8 + 3 = -9$. Treating all moves as left ($4-6-9-7-8-3$) gives -29 -type errors; mis-counting the rights as one more left lands at -15 ; sign flip gives $+9$.

13. (A) Inside brackets: $(-5) + (-2) - (-4) = -5 - 2 + 4 = -3$. Then $(-3) \times (-3) = +9$; $+9 + (-6) = +3$. Reading $-(-4)$ as -4 makes the bracket -11 and the answer -33 -ish; forgetting to add the trailing -6 gives $+9$; sign error on the product gives -9 .

14. (D) A product is negative exactly when an odd number of its factors are negative; this is independent of how many factors there are in total. So the count is odd (could be 1, 3, 5, or 7) — 'even' would force a positive product, and 'exactly seven' / 'exactly one' are only particular odd cases.

15. (B) Do $\div$ and $\times$ first: $(-84) \div (-7) = +12$; $(-5) \times (-3) = +15$; $(-2) \times (+4) = -8$. Now $12 + 15 - (-8) = 12 + 15 + 8 = 35$. Reading $-(-8)$ as -8 gives 19; sign-slipping the $+15$ gives a smaller value; dropping a term gives 27.

16. (C) 'How much greater' is the difference $(-13) - (-41) = -13 + 41 = 28$. Subtracting the wrong way $(-41) - (-13)$ gives -28 ; adding the magnitudes gives 54; both sign-and-order slips give -54 .

17. (C) $(-2)^4 = +16$ (four negatives, even); $(-2)^3 = -8$ (odd); $(-2)^2 = +4$. So $16 + (-8) - (+4) = 16 - 8 - 4 = 4$. Treating $(-2)^4$ as -16 or $(-2)^3$ as $+8$ flips terms; mishandling the final $- (+4)$ as $+4$ gives 28.

18. (A) Do $\div$ and $\times$ left to right: $48 \div (-6) = -8$; $-8 \times 2 = -16$; the term is $-(-16) = +16$. So $30 + 16 + (-5) = 30 + 16 - 5 = 41$. Subtracting 16 instead of adding it (sign slip on $\div$ by -6) gives 9; doing the subtraction before $\times/\div$ corrupts the rest.

19. (A) $a + 0 = a$ (additive identity) and $a \times 1 = a$ (multiplicative identity) hold for every integer. Multiplying by 0 gives 0, not the integer back; 0 IS the additive identity, so the last claim is false; the first option swaps the two identities.

20. (A) Magnitude: $1 \times 2 \times 3 \times 4 \times 5 \times 6 = 720$. Six negative factors — an even count — give a positive product, so $+720$. An odd-count assumption gives -720 ; adding the magnitudes gives 21.

21. (B) Total rise = $(-60) - (-540) = -60 + 540 = +480$ m over 6 steps, so $480 \div 6 = +80$ m each (a rise is positive). Dividing $540 \div 6 = 90$ ignores that it stops at -60 ; dividing $600 \div 6$ over-counts; a rise cannot be negative, ruling out -80 .

22. (B) Two factors of -1 multiply to $+1$, leaving $n \times n = n^2$. So the value is n^2 , which is ≥ 0 for every integer (e.g. $n = -3$ gives 9). It is never $-n^2$; 'always negative' is false since $n^2 \geq 0$; $2n$ confuses multiplication with addition.

23. (D) Seven copies of -12 : $7 \times (-12) = -84$. Counting six copies gives -72 ; counting eight gives -96 ; dropping the overall negative sign gives $+84$.

24. (B) Brackets first: $(-4) + (-1) = -5$. Then $(-90) \div (-5) = +18$; $+18 - (-6) = +18 + 6 = +24$. Stopping at $+18$ forgets the $-(-6)$; reading $-(-6)$ as -6 gives $+12$; a sign error on the division gives -24 .

25. (D) Multiplying a single factor across a sum, $a \times (b + c) = a \times b + a \times c$, is exactly the distributive property. Associativity regroups three multiplied factors; commutativity of addition only reorders the summands; the additive-inverse property is about $a + (-a) = 0$.

26. (A) Left to right for $\times/\div$: $(-144) \div 6 = -24$; $-24 \times (-1) = +24$; $+24 \div (-3) = -8$; then $-8 + (-4) = -12$. Dropping the final $+(-4)$ gives -8 (then mis-rounded); a sign slip on the

$\div(-3)$ gives +12; mishandling two negatives gives -20.

27. (C) Correct: $12 \times 5 = 60$. Wrong: $5 \times (-3) = -15$. Unattempted: $3 \times (-1) = -3$. Total: $60 - 15 - 3 = 42$. Ignoring the unattempted penalty gives 45; ignoring both penalties gives 60; only counting the wrong penalty gives 45/48 variants.

28. (D) $(-7) \times x + 11 = -31 \Rightarrow (-7)x = -42 \Rightarrow x = (-42) \div (-7) = 6$. Forgetting to subtract 11 first gives $(-7)x = -31$; a sign error in the final division (negative $\div$ negative) gives -6; using $+42 \div (-7)$ gives -6 as well.

29. (C) Inside brackets: $(-8) - (-3) + (-2) = -8 + 3 - 2 = -7$. Then $(-5) - (-7) = -5 + 7 = +2$. Reading the outer $-(-7)$ as -7 gives -12; sign-slipping the bracket to +7 gives -12 the other way; mis-adding inside gives -2.

30. (D) First bracket: $(-6) - (-10) = -6 + 10 = +4$. Second bracket: $(-3) - (+5) = -8$. Product: $(+4) \times (-8) = -32$. Reading the first bracket as -16 (sign error) or the product sign as positive gives +32; mis-multiplying gives ± 16 .

31. (C) An integer plus its opposite is 0, so we compute $(-29) - 0 = -29$. Confusing the zero result with the final answer gives 0; flipping the sign gives +29; doubling -29 gives -58.

32. (D) Evaluate each: $(-3) \times (-2) = +6$; $(-15) \div 3 = -5$; $(-4) + (-4) = -8$; $6 - 9 = -3$. Ascending order of values -8, -5, -3, +6 corresponds to $(-4) + (-4)$, $(-15) \div 3$, $6 - 9$, $(-3) \times (-2)$. Swapping -8 and -5 (treating -5 as smaller than -8) gives the first distractor; reversing gives the third.

33. (B) Multiply first: $(-4) \times (-4) = +16$ and $4 \times (-4) = -16$. Expression: $16 - (-4) - (-16) = 16 + 4 + 16 = 36$. Reading $-(-16)$ as -16 gives 4; reading $-(-4)$ as -4 gives 28; combining both sign slips gives -4.

34. (A) $p \times q \times r$ is a product of three negatives, hence negative. $p + q + r$ is a sum of three negatives, hence negative. Negative $\times$ negative = positive, so the whole expression is positive for all such p, q, r . The result is fully determined.

35. (C) Brackets: $14 - 30 = -16$; $(-16) \times (-3) = +48$. Division: $(-56) \div (-7) = +8$. Sum: $48 + 8 = 56$. A sign error on the product gives $-48 \rightarrow -40$; mishandling the negative $\div$ negative gives 40; both slips give -56.

36. (D) There are five negative factors; an odd number of negatives makes the product negative. (No factor is 0, so it is non-zero.) The actual value is -3840, but the question only needs the sign, which is negative; an even-count assumption gives 'positive'.

37. (D) $(-96) \div x = +8 \Rightarrow x = (-96) \div (+8) = -12$ (negative $\div$ positive = negative, and $-12 \times 8 = -96$ ✓). Choosing +12 gives $(-96) \div 12 = -8$, the wrong sign; subtracting instead of

dividing gives ± 88 .

38. (B) $98 = 100 + (-2)$, so $(-6) \times 98 = (-6) \times 100 + (-6) \times (-2) = -600 + 12 = -588$. Writing 98 as $100 - 2$ needs $(-6) \times 100 - (-6) \times 2 = -600 + 12 = -588$ too, but the distractors mis-state the arithmetic as -612 ; using $(-6) \times (-2) = -12$ is the classic two-negatives sign error.

39. (A) $x \times (-7) = +56 \Rightarrow x = (+56) \div (-7) = -8$ (and $(-8) \times (-7) = +56$ ✓). The product is positive with one factor negative, so the other must also be negative; $+8$ would give $(+8) \times (-7) = -56$; multiplying instead of dividing gives ± 392 .

40. (C) Net change: $-250 + 180 - 90 + 60 = -100$ m. Three times that is $3 \times (-100) = -300$ m. Forgetting to triple gives -100 ; dropping the sign before tripling gives $+300$; tripling the magnitude sum (580) gives -580 .

41. (C) Decreasing means each term is less than the one before: $7 > 2 > 0 > -3 > -8 > -15$ holds. The first distractor is ascending; the others break order at the $0/-3$ swap and the $-15/-8$ swap, where $-15 < -8$ is misjudged.

42. (D) First bracket: $(-3) - (-8) = -3 + 8 = +5$, so $(-7) \times 5 = -35$. Second term: $(-7) \times (-4) = +28$, subtracted: $-(+28) = -28$. Total: $-35 - 28 = -63$. Reading the bracket as -5 flips the first term; mishandling $-(+28)$ as $+28$ gives $+7$ -type values.

43. (A) Adding, subtracting, or multiplying two integers always yields an integer, so closure holds for those three. Division can leave the integers (e.g. $(-7) \div 2$ is not an integer), so they are NOT closed under division. Subtraction IS closed, ruling out the third option.

44. (B) $(-2)^3 = -8$; $(-3)^2 = +9$; product $= -72$. $(-4)^2 = +16$; $+16 \div (-2) = -8$; subtracted: $-(-8) = +8$. Total: $-72 + 8 = -64$. Reading $(-2)^3$ as $+8$ or dropping the final $+8$ gives $-80/-72$; a full sign flip gives $+64$.

45. (D) Need sum -7 , product $+12$. A positive product with a negative sum means both are negative: $(-3) + (-4) = -7$ and $(-3) \times (-4) = +12$ ✓. 3 and 4 give sum $+7$; $(-6) + (-1) = -7$ but product $+6$; $(-12) + (-1) = -13$.

46. (D) Rewrite as adding opposites: $-13 + 21 - 8 + 6 - 4 = (-13 + 21) = 8$; $8 - 8 = 0$; $0 + 6 = 6$; $6 - 4 = 2$. Misreading one $-(-)$ as a subtraction flips a sign to give -2 ; adding all magnitudes gives -52 -type errors.

47. (B) Evaluate: $(-2)^5 = -32$ (five negatives, odd $\rightarrow$ negative); $(-3)^3 = -27$; $(-8) - (-2) = -6$; $(-10) + (-4) = -14$. The most negative is -32 . A common slip treats $(-2)^5$ as $+32$ or -10 , hiding that it is the smallest.

48. (B) Division left to right: $(-128) \div (-8) = +16$; $+16 \div (-2) = -8$. Multiplication: $(-3) \times (-2) = +6$, subtracted: $-(+6) = -6$. Total: $-8 - 6 = -14$. A sign error making the division $+8$ gives $+2$; reading $-(+6)$ as $+6$ gives -2 ; both slips give $+14/+22$.

49. (B) Debits are negative, credits positive: $-320 + 185 - 95 + 430 = (-320 - 95) + (185 + 430) = -415 + 615 = +200$. Swapping debit/credit signs gives -200 ; adding all magnitudes gives 1030; mis-pairing one term gives -230 .

50. (C) $\times/\div$ first: $6 \times (-4) = -24$, so $-(-24) = +24$; $(-48) \div (-8) = +6$. Now $(-9) + 24 + 6 - 5 = 16$. Reading $-6 \times (-4)$ as -24 (subtracting) gives a much smaller value; a sign slip on the division gives 10/22 variants.

51. (C) Signed moves from $+5$: $5 - 9 + 4 - 7 + 2 - 3 = (5-9)=-4$; $-4+4=0$; $0-7=-7$; $-7+2=-5$; $-5-3=-8$. Treating every move as down gives -33 -type errors; mis-counting an up as down lands at -18 ; sign flip gives $+8$.

52. (A) Compute each product: $(-15) \times (-4) = +60$; $(-15) \times (+2) = -30$, and it is subtracted, so $-(-30) = +30$; $(-15) \times (-1) = +15$. Total: $60 + 30 + 15 = 105$. Treating the middle term as -30 instead of $+30$ (forgetting the leading minus turns it positive) gives 75; further sign slips give 45; a full sign reversal gives -105 .

53. (C) -7^2 means $-(7^2) = -49$; $(-7)^2 = +49$; $(-1)^3 = -1$. Sum: $-49 + 49 + (-1) = -1$. Reading -7^2 as $(-7)^2 = +49$ gives $+97$; dropping the cube term gives $0/+1$; sign-flipping both squares gives -97 .

54. (A) First bracket: $(-4) \times (-3) = +12$; $12 - 2 = 10$. Second bracket: $(-2) + (-1) - (-4) = -2 - 1 + 4 = +1$. Product: $10 \times 1 = 10$. Reading the second bracket as -1 (sign error on $-(-4)$) gives -10 ; mis-multiplying gives ± 30 .

55. (C) Factor (-6) : $(-6) \times [(-5) + (-3) - (+2)] = (-6) \times (-10) = +60$. Directly: $(-6)(-5)=30$; $(-6)(-3)=18$; $(-6)(2)=-12$, subtracted is $+12$; $30+18+12 = 60$. A sign error making the last term -12 gives $48/36$; full sign flip gives -60 .

56. (B) Any product containing a factor of 0 is 0, regardless of how many negative factors are present — the zero-product property overrides the sign-parity rule. So it is 0, not negative or positive, and the count of negatives is irrelevant.

57. (D) Net change: $-2 - 2 + 5 - 3 - 2 + 1 = (-2-2-3-2) + (5+1) = -9 + 6 = -3$. Final = $4 + (-3) = +1$ °C. Forgetting the starting 4 gives -3 ; mis-signing one fall gives $+7$; a single arithmetic slip gives -1 .

58. (D) Innermost bracket: $(-6) + (-2) = -8$. Next: $(-4) - (-8) = -4 + 8 = +4$. Outermost: $(-10) - (+4) = -14$. Reading the inner $(-4) - (-8)$ as $(-4) - 8 = -12$ then $(-10) - (-12) = +2$ gives the $-2/+2$ traps; mishandling the outer $-(+4)$ as $+4$ gives -6 .

59. (B) Positive factors never change a product's sign; an even count of negative factors contributes a positive sign. With no zero present, the product is non-zero and positive. An odd count would be needed for negative; 'even' confuses the parity of the count with the sign of the value.

60. (A) First bracket: $(-7) \times (-2) = +14$; $14 - 8 = 6$. Second bracket: $(-1) + (-2) = -3$. Now $6 \div (-3) \times (-3)$, left to right: $6 \div (-3) = -2$; $-2 \times (-3) = +6$. Stopping at -2 ignores the $\times(-3)$; a sign slip on either step gives $-6/+18$.